AMARNATH S. PATEL

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EDUCATION

University of Central Florida — **Undergraduate Student** — **4.00 GPA**
B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor — August 2025 – Expected May 2029

Florida Atlantic University — **University coursework via FAU High School, ages 14–18** — **3.66 GPA**
111 credit hours completed between ages 14 and 18, alongside high school — August 2021 – May 2025

RESEARCH EXPERIENCE

Undergraduate Researcher — UCF Astrophotonics Lab, CREOL — August 2025 – Present
Supervisor: Dr. Stephen Eikenberry

- Measuring the wavelength-dependent complex transfer matrix of a photonic lantern by off-axis digital holography, extending Dobias et al., *Opt. Express* **34**(9), 17217 (2026). A photonic lantern couples a multimode fiber to an array of single-mode fibers; using one to feed starlight into single-mode photonic devices requires knowing exactly how light entering each port emerges from the multimode end, in amplitude and phase, at every wavelength of interest.
- Align the interferometric bench and acquire holograms of the multimode output for each input port across the C-band (1525–1575 nm), recovering full amplitude and phase — one row of the transfer matrix per measurement.
- Implemented the phase-retrieval and mode-decomposition analysis: FFT sideband isolation and demodulation, Butterworth low-pass filtering, joint numerical optimization of mode-field diameter, defocus quadratic phase and field position, then decomposition onto the LP basis to recover complex modal amplitude and phase.
- Produced the per-port reconstruction-fidelity characterization for 6- and 7-port lanterns, reaching ≈98% fidelity $|\langle E_{\mathrm{rec}}, E_{\mathrm{sim}}\rangle|^2$ against simulated fields.
- Automated the four-instrument acquisition chain — tunable laser, InGaAs camera, fiber switch and motorized polarization control — so a complete all-port × C-band sweep runs unattended, with polarization optimized in-loop for peak fringe contrast and saturated frames rejected on arrival. A full dataset went from days of manual bench time to a single hands-off run.
- Own the telescope-node software for PolyOculus, an instrument program building photometric observing capability from a networked array of 8 telescopes, aimed at a redshift-drift measurement. Wrote centroid-based autoguiding from the guide camera to INDI mount guide pulses (simulated tracking error 3.01″ ⟶ 0.29″ RMS) and closed-loop piezo steering-mirror control holding a laser spot to 0.2 px, and brought the array's Jetson Nanos up on mainline Linux to replace NVIDIA's end-of-life image, with every node rebuildable from one repository.

Independent Research — CELERIS: computational electromagnetics and metalens design — 2026

- Built a from-scratch solver for Maxwell's equations in periodic subwavelength structures via rigorous coupled-wave analysis / the Fourier modal method: 1D TE/TM multilayer gratings with Li and Liu–Fan inverse-rule factorization, the full 2D-vectorial $P \cdot Q$ formulation for biperiodic nanopillar arrays, and stable Redheffer scattering-matrix layer recursion.
- Validated every layer of the stack against closed-form physics, an independent method, or energy conservation: RCWA versus the transfer-matrix method to 1e-6, multilayer S-matrix to 1e-15, 2D⟶ 1D degenerate reduction to 5e-12, Σ(diffraction efficiency) = 1.000000, and diffraction-limited designed focal spots at FWHM = $\lambda f/D$. Independently cross-validated against the grcwa and Stanford S⁴ solvers (≈ 1e-7) and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Implemented the physical design and analysis capability standalone RCWA kernels lack: Sellmeier and tabulated n, k dispersion, meta-atom library sweeps with automatic etch-depth selection for full-2π phase coverage, gradient-based (Adam) inverse design, and Pancharatnam–Berry geometric-phase and polarization-multiplexed lenses exploiting form birefringence. Analysis covers Strehl ratio, Zernike wavefront, MTF, chromatic focal shift, through-focus caustic and fabrication-tolerance Monte Carlo, with fabrication-ready GDSII export.
- Accelerated Rayleigh–Sommerfeld far-field propagation as a CUDA kernel, benchmarked against the project's own optimized 16-core CPU path: 4.8–5.8× at 92k pillars, agreeing to machine precision. C++23/CUDA with Python bindings. *Sole-author manuscript in preparation (Computer Physics Communications).*

Independent Research — Stardust: archival recovery of newly discovered near-Earth asteroids — October 2026

- Began at Knight Hacks IX as a team project; I built the orbit and detection pipeline. From only the sightings that existed 4 days after discovery, a batched GPU N-body integrator (PyTorch; DE440 perturbers plus solar relativity) samples the orbit uncertainty, projects it onto years of pre-discovery ZTF images, and searches them with a U-Net detector trained on 4,000 real cutouts and a classic streak filter.
- Blind backtest on 72 asteroids once on NASA's Sentry impact-risk list: 167 verified, never-connected sightings of 22, tightening their orbits up to 52× (median 7.1×); a frozen held-out round recovered 16 of 50. Validated against JPL on 2024 YR4 (2032 Earth miss 279,443 km versus JPL's 278,333 km). Live run on NASA's current risk list found 202 pre-discovery sightings of 23 asteroids, confirmed by both detectors.

Undergraduate Research Assistant — UCF Physics Department — March 2026 – Present
Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Research on automated assessment generation for introductory physics instruction: built ESTELA, which produces multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, and an agentic generation–evaluation pipeline that iteratively drafts and validates new problems against instructor criteria.
- Materials authored for the project are in use by external adopters following a faculty workshop. Manuscript in preparation.

Undergraduate Researcher — FAU Grant-Funded AI Safety Research Project — January 2024 – March 2025
Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10 statistical coherence and detection measures (GPT-2 perplexity, RoBERTa, BERT next-sentence prediction, LSA, NLI, burstiness) on a 255,000-passage corpus; identified GPT-2 perplexity as the strongest discriminator (3.35× separation, 17.5 versus 58.5) and showed BERT NSP to be uninformative for the task.
- Built four adversarial evasion pipelines (iterative rewriting, tree-search decoding, list-branching, reinforcement learning) over 3,125 generated sequences and characterized a consistent quality–evasion tradeoff. Grant-funded with HPC access.

PRESENTATIONS

Batch Processing for Automated Grading via Azure OpenAI June 2026
University of Central Florida, Downtown Campus. ESTELA project, Physics Education Research.

Coherence and Detection Approaches for Identifying AI-Generated Text March 2025
Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

TECHNICAL REPORTS

Coherence and Detection Approaches for Identifying AI-Generated Text 2025
Empirical study of AI-text detection and adversarial evasion: benchmarked 10 coherence and detection methods and built four evasion pipelines, characterizing a consistent quality–evasion tradeoff. Grant-funded, Florida Atlantic University.

TECHNICAL SKILLS

Instrumentation: interferometric alignment, GPIB/VISA and RS-232 instrument control, INDI, GigE Vision camera acquisition, tunable IR laser control, motorized polarization control, fiber coupling, FITS and GDSII data formats

Scientific computing: FFT and signal processing, phase retrieval and modal decomposition, electromagnetic simulation (RCWA), gradient-based optimization, Monte-Carlo tolerance analysis, GPU/CUDA acceleration, NumPy, SciPy

Programming: C/C++ (C++23), CUDA, Python, Rust, Go, shell (Fish, Bash), LaTeX/Typst

Tools: Linux, Git, CMake, Docker, Qt/PySide6

RELEVANT COURSEWORK

University of Central Florida — *Physics:* Geometric Optics & Lab, Modern Physics, Quantum Information Processing, Mathematical Methods for Physics, Independent Research [PHY 4912], Electricity & Magnetism I (in progress). *Mathematics:* Applied Linear Algebra, Complex Analysis, Linear Algebra (proof-based, in progress), Partial Differential Equations (in progress). *Computer science:* C Programming, Discrete Structures, Object-Oriented Programming.

Florida Atlantic University — General Physics I and II (honors), Calculus I–III (honors), Ordinary Differential Equations, Elementary Matrix Algebra, Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Deep Learning, C++ Programming.

HONORS & AWARDS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) 2025
Lockheed Martin Award, "Highest Level of Engineering Excellence," AEV Competition 2024
2nd Place in Division, Advanced Experimental Vehicles Competition 2024

SERVICE & OUTREACH

Secretary, National Society of Black Physicists — UCF chapter 2025 – Present
Teaching Assistant, Calculus — office hours, exam review and grading for 70 undergraduates 2024 – 2025
Founder and President, Coding Club, FAU High School — now an official Hack Club chapter 2023 – 2025